

Titanic Survival Analysis — Executive Summary

Data-driven insights on survival patterns and risk segmentation (n = 891)

Key Insight 1: Gender is the strongest predictor

Female passengers had a significantly higher survival rate (74%) compared to males (19%). Statistical testing confirms this gap is highly significant ($p < 0.001$), making gender the most dominant factor influencing survival outcomes.

Key Insight 2: Socioeconomic class strongly impacts survival

Passenger class shows a clear gradient in survival rates: 1st Class (63%), 2nd Class (47%), and 3rd Class (24%). This demonstrates that access to resources and priority significantly improved survival likelihood.

Key Insight 3: Intersection of class and gender amplifies outcomes

The highest survival group was 1st-Class women ($\approx 97\%$), while 3rd-Class men had the lowest survival ($< 20\%$). This highlights how multiple factors combine to create extreme differences in outcomes.

Business Interpretation

The findings mirror modern systems where premium access leads to better outcomes—not just perception. In business terms, resource allocation, prioritization, and segmentation directly influence success rates.

Recommendation

Organizations should treat key variables such as customer tier and demographic segments as independent drivers of outcomes. Strategies must ensure that premium offerings deliver measurable value, while also minimizing disparities in lower segments.